KARTHIK GOUD KASULA

Data Engineer - Snowflake · Databricks · PySpark · Airflow · dbt
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PROFESSIONAL SUMMARY

Data Engineer with 3 years of experience designing, developing and optimizing batch and streaming ETL/ELT pipelines using Python, SQL and PySpark across Azure Databricks and Snowflake, including recent experience supporting a major U.S. bank. Experienced in real-time payment processing using Spark Structured Streaming and Delta Lake to deliver intraday visibility within a 5-minute window and in building dbt-based dimensional models with Star Schema design, SCD Type 2 history and PII masking. Improved PySpark job performance by reducing runtime from 60 to 35 minutes through data skew mitigation and Delta file compaction. Skilled in orchestrating workflows with Airflow and implementing Great Expectations data quality checks and dbt reconciliation tests to ensure reliable financial reporting.

TECHNICAL SKILLS

Languages: SQL, Python
Processing: Apache Spark (PySpark), Spark SQL, Kafka
Orchestration / transform: Apache Airflow, dbt, Azure Data Factory
Cloud-Azure, AWS: Azure Databricks, ADLS Gen2, Azure Synapse Analytics, Event Hubs
Warehouse / lakehouse: Snowflake, Delta Lake, dimensional modeling, SCD Type 2
Quality and DevOps: dbt tests / Great Expectations, Git, GitHub Actions, Terraform
BI and consumers: Power BI, Tableau

PROFESSIONAL EXPERIENCE

Truist (Contract via DeepData Inc) **May 2025 – Present**
Data Engineer Remote,TX

- Engineered incremental PySpark pipelines in Azure Databricks to capture customer and account changes using last-modified timestamps, watermarks, and lookback windows. Implemented Delta Lake MERGE upserts and batch-level audit columns across six source tables, replacing nightly full reloads.
- Developed Spark Structured Streaming jobs in Azure Databricks to consume payment transaction events from Azure Event Hubs through its Kafka-compatible endpoint. Managed late-arriving events using event-time watermarks and wrote checkpointed micro-batches to Delta Lake, delivering intraday payment visibility within a 5 minutes processing window.
- Modeled customer, account, and transaction data into star-schema fact and dimension tables in Snowflake using dbt. Implemented snapshots for SCD Type 2 account history and role-based masking policies for PII columns, enabling finance analysts to perform point-in-time balance analysis while protecting sensitive data.
- Implemented automated data quality and reconciliation controls using Great Expectations and dbt tests to validate record counts and debit/credit control totals against source data by business date. Routed failures to exception tables and prevented downstream financial reports from refreshing until discrepancies were resolved.
- Optimized long-running PySpark and Spark SQL jobs by analyzing Spark UI stage metrics, reducing processing runtime from 60 to 35 minutes. Applied broadcast joins for small reference tables, mitigated data skew on high-volume account keys, and compacted small Delta files using OPTIMIZE and Z-ORDER.
- Orchestrated 12 Apache Airflow DAGs to manage Azure Databricks jobs and dbt transformations, implementing retries, SLA monitoring, failure alerts, and idempotent partition-level reruns. Deployed pipeline changes through GitHub Actions CI/CD and collaborated with QA and senior engineers to investigate and resolve production incidents.

TechSolutions Inc. **Aug 2022 – Dec 2023**
Junior Data Engineer Hyderabad, India

- Automated ingestion from five paginated REST API endpoints and vendor-provided JSON/CSV feeds using Python, flattening nested structures into standardized Parquet files in Azure Data Lake Storage Gen2 (ADLS Gen2). Implemented retry handling for request timeouts and API rate limits under senior engineer code review.
- Built parameterized Azure Data Factory (ADF) pipelines using Copy activities and scheduled triggers to perform daily full-refresh loads of 10 relational source tables into ADLS Gen2 and Azure Synapse Analytics staging tables, following established team templates and naming standards.

- Transformed staged data in Azure Synapse Analytics using SQL joins, aggregations, and window functions to develop reporting views for Power BI dashboards. Translated analyst-defined business rules into documented, peer-reviewed SQL queries to support consistent reporting and downstream analytics.
- Validated pipeline outputs using reusable SQL checks for source-to-target row counts, null keys, duplicate records, and data type mismatches. Documented validation results and escalated mapping defects to senior engineers for resolution before user acceptance testing (UAT).
- Migrated a single-node Python data-cleansing script to PySpark notebooks in Azure Databricks as growing daily file volumes exceeded the existing batch-processing window. Implemented schema casting, deduplication, and timestamp normalization, reducing processing time from 40 to 15 minutes.
- Troubleshot failed Azure Data Factory (ADF) pipeline runs during production support rotations, identifying issues related to schema drift, expired credentials, and delayed source files. Implemented fixes through Git feature branches and pull requests and documented resolutions in team runbooks to support consistent incident resolution.

TECHNICAL PROJECTS

Real-Time Transaction Streaming Pipeline

github.com/Karthik-Goud-Kasula/Real-Time-Transaction-Streaming-Pipeline

Apache Kafka, Spark Structured Streaming, Delta Lake, Databricks, Python

- Built a real-time data pipeline using Kafka, Spark Structured Streaming, and Delta Lake to process approximately 250 payment events per second. Organized data into Bronze, Silver, and Gold layers, using event-time watermarking and event-ID deduplication to handle late-arriving and duplicate events. Implemented idempotent for each Batch MERGE operations to prevent duplicate writes and ensure reliable processing.
- Refreshed gold-layer fraud and volume aggregates in under a minute, with dead-letter routing for malformed events and OPTIMIZE with ZORDER to keep queries fast as Delta tables grew.

Analytics Engineering Layer on Snowflake

github.com/Karthik-Goud-Kasula/Analytics-Engineering-Layer-on-Snowflake

dbt, Snowflake, Apache Airflow, GitHub Actions

- Transformed the Kaggle Financial Sample dataset into 20+ dbt models across staging, intermediate, and mart layers, implementing incremental MERGE strategies, SCD Type 2 snapshots, and automated tests for uniqueness, not-null, relationships, and accepted values across the data pipeline.
- Set up Slim CI in GitHub Actions (state:modified+) to build and test only changed models on each pull request, and published dbt docs with column-level lineage.

EDUCATION

University of North Texas — **Jan 2024 – May 2025**
M.S. in Advanced Data Analytics — Denton, USA

Vidya Jyothi Institute of Technology — **Aug 2019 – Apr 2023**
B.Tech, Electronics and Communication Engineering — Hyderabad, India

CERTIFICATIONS

Microsoft Certified: Fabric Data Engineer Associate (DP-700) — March 2026